Understanding Customer Behaviour and Revenue Drivers in Electronic Sales: The Role of Gender, Age, Pricing, and Add-ons

Before reading the full text, you could explore this webpage I created for the assignment. I hope it will enhance your interest for my work:

https://69cc08dd780d5c7eb6f564a1--advanced-individual-assignment.netlify.app

Introduction

Based on an initial examination of the dataset and its key features, the dataset contains 20,000 observations and 16 variables: *Customer ID, Age, Gender, Loyalty Member, Product Type, SKU, Rating, Order Status, Payment Method, Total Price, Unit Price, Quantity, Purchase Date, Shipping Type, Add-ons Purchased,* and *Add-on Total*. The data covers one year from September 2023 to September 2024, providing a comprehensive record of customer purchase behaviour over time.

After reviewing the dataset and completing data cleaning, three research questions were developed based on the data's characteristics and an interest in understanding customer purchasing patterns. The analysis is structurally arranged around two more focused sub-questions and one main research question.

The first study question investigates if consumer behaviour patterns, such as loyalty conversion, purchase cancellation, payment method selection, and delivery preferences, differ by gender and age:

RQ1: Do gender and age differences exist in loyalty conversion, cancellation, payment method, and shipping preferences?

The second research question investigates whether gender or age group is linked to distinct price tier choices within the same product category, extending the demographic comparison to price sensitivity:

RQ2: Do gender or age group predict price tier preferences within the same product category?

Lastly, the final study question discusses the role of add-on purchases and the factors that influence overall spending. After analysing the overall distribution of *Total Price* and the contributions of *Product Type* and *Quantity*, it looks at whether consumers who buy less expensive product models behave differently when making add-on purchases than those who buy more expensive models:

RQ3: What factors drive total spending, and does product model price level influence add-on purchasing behaviour?

When combined, these enquiries offer a methodical investigation of consumer behaviour. RQ1 and RQ2 examine whether the client base is meaningfully divided by demographic traits. RQ3 then determines the structural drivers of revenue.

Analysis

During data cleaning, it was observed that the *Payment Method* variable contained two distinct spellings: *PayPal* and *Paypal.* These were standardised by converting all entries to *PayPal* to ensure consistency in the dataset.

After removing observations with missing gender information, the dataset contained 19,999 records. Among these observations, there are 12,135 unique customers.

Table 1 Descriptive Statistics for Numeric Variables

index	Age	Total Price	Unit Price	Quantity	Add-on Total
count	19999	19999	19999	19999	19999
mean	49	3180.26	578.63	5.49	62.25
std	18	2544.98	312.28	2.87	58.06
min	18	20.75	20.75	1	0.00
25%	33	1139.68	361.18	3	7.62
50%	49	2534.49	463.96	5	51.70
75%	65	4639.6	791.19	8	93.84
max	80	11396.8	1139.68	10	292.77

According to *Table 1*, the *Age* range is very wide, spanning from 18 to 80 years old, with both the median and mean equal to 49. *Total Price* also shows a very large range, from $20.75 to $11,396.80. The *Unit Price* ranges from $20.75 to $1,139.68, which reflects the wide variety of electronic products across different price tiers. The *Add-on Total* ranges from $0 to $292.77, with a median of $51.70.

RQ1 Demographic Differences in Consumer Behaviour
RQ1A Gender Analysis

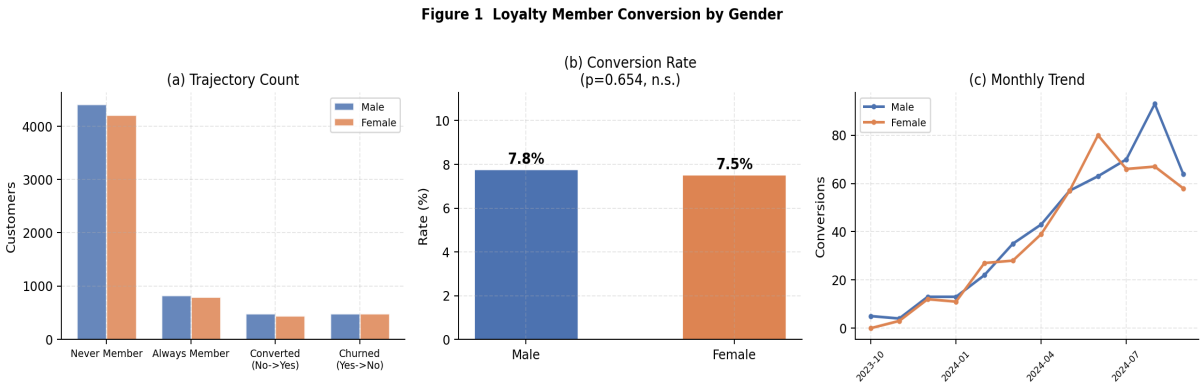

Figure 1 Loyalty Member Conversion by Gender

According to *Figure 1*, the *Never Member* group accounts for the largest proportion in both gender groups, each exceeding 4,000 customers. The number of *Converted* customers (from non-member to member) is almost equal to the number of *Churned* customers (from member to non-member). This balance may suggest that the company is not doing enough to retain its existing loyalty members.

The loyalty member conversion rate does not show a meaningful gender difference. The conversion rate for male customers is 7.8%, compared with 7.5% for female customers. The chi-square test shows no statistically significant difference ($p = 0.654$). This finding is consistent with prior research suggesting that gender differences in online loyalty behaviours are generally small and may vary substantially by context (Kanwal et al., 2022).

According to the monthly conversion trend shown in *Figure 1(c)*, female customer conversions reached their peak in June 2024, while male customer conversions peaked in August 2024. One possible explanation is that the company launched a marketing campaign during that period, but its effect on male customers was observed with a lag of approximately two months.

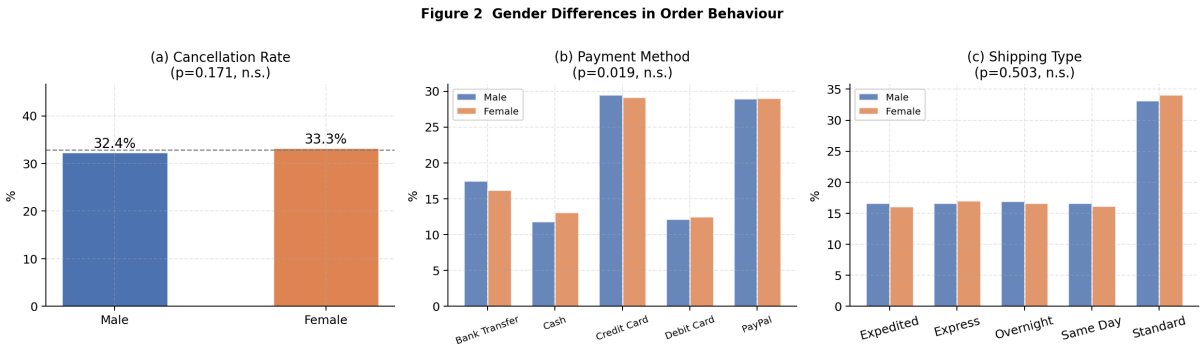

Figure 2 Gender Differences in Order Behaviour

According to *Figure 2*, gender does not show a significant difference across any of the three behavioural metrics. All chi-square tests show p-values above 0.05, indicating that gender does not significantly affect order cancellation, payment method preference, or shipping type selection.

In terms of absolute values, the cancellation rate exceeds 30% for both groups, which is a relatively high figure. One possible explanation is that slow shipping leads customers to abandon orders. Regarding payment methods, *Credit Card* and *PayPal* together account for the majority of transactions, each with a share roughly double that of *Debit Card*. For shipping types, *Standard* shipping has by far the largest share, approximately twice as high as any other type, suggesting that customers are generally not willing to pay a premium for faster delivery.

Table 2 RQ1A Statistical Summary

Metric	Male (%)	Female (%)	Chi-square (χ^2)	p-value
Loyalty Conversion (No→Yes)	7.8	7.5	0.20	0.654
Order Cancellation	33.3	32.4	1.88	0.171
Credit Card Usage	29.5	29.2	0.20	0.658
Shipping Type (overall)	-	-	3.34	0.503

As shown in *Table 2*, none of the four behavioural indicators shows a statistically significant gender difference. This suggests that the company may not currently be implementing effective customer segmentation strategies.

RQ1B Age Group Analysis

To test whether age group predicts any of the same behavioural indicators, chi-square tests were conducted for each RQ1 metric across four age brackets (18-30, 31-45, 46-60, 61-80). Customers were divided into age groups based on quartile-informed boundaries, yielding four groups of broadly comparable size.

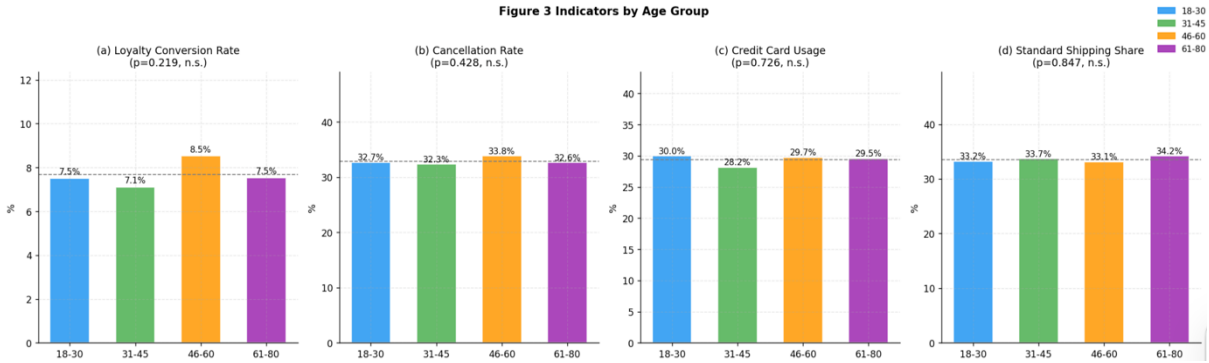

Figure 3 Behavioural Indicators by Age Group

Figure 3 shows the four RQ1 indicators broken down by age group. The dashed horizontal line in each panel marks the overall average, and the four bars are visually indistinguishable across groups. *Table 3* presents the statistical results.

Table 3 Age Group Statistical Summary

Metric	18-30 (%)	31-45 (%)	46-60 (%)	61-80 (%)	Chi-square (χ^2)	p-value
Loyalty Conversion	7.5	7.1	8.5	7.5	4.429	0.219
Order Cancellation	32.7	32.3	33.8	32.6	2.772	0.428
Credit Card Usage	30.0	28.2	29.7	29.5	8.728	0.726
Standard Shipping	33.2	33.7	33.1	34.2	7.157	0.847

None of the four indicators is significantly associated with age group. Loyalty conversion rates range from 7.1% to 8.5%, with p = 0.219. Cancellation rates range from 32.3% to 33.8%, with p = 0.428. Credit card usage ranges from 28.2% to 30.0% across groups (p = 0.726). Standard shipping preference is consistent at approximately 33 - 34% across all age brackets (p = 0.847). Cohen's w values are all below 0.03 for each test, indicating negligible effect sizes.

The convergent finding from RQ1A and RQ1B is that neither *Gender* nor *Age Group* explains variation in loyalty conversion, order cancellation, payment method preference, or shipping type selection. Customer behaviour across these dimensions is largely homogeneous regardless of the demographic variable examined.

RQ2 Demographic Differences in Price Tier Preference

RQ2A Why Quantile-Based Binning Failed

The initial approach to assigning price tiers used a quantile-based binning method (pd.qcut(x, q=3)), which is designed for continuous variables with many unique values. It divides the data into three groups of equal frequency by computing quantile-based cut points.

However, the *Unit Price* variable in this dataset is not continuous. It is a discrete categorical variable determined entirely by the *SKU* purchased, with each product category containing only 2 to 4 distinct price points. In addition, one *SKU* typically accounts for approximately 50% or more of all orders within a category.

As a result, the 33rd and 67th percentile cut points often coincide with the same price value, leading to non-unique bin edges. This prevents the quantile-based binning method from forming three distinct groups. When duplicate edges are removed, the number of resulting bins no longer matches the intended three labels (Low, Mid, High), further invalidating the classification. In practice, this leads to degenerate outputs in which all observations are

assigned to a single category, rendering the approach analytically uninformative. Anomaly *SKU* with fewer than 10 orders are excluded, and finally, I got 19,995 observations. *Table 4* shows the full *SKU* inventory and *Table 5* shows the summary of different products with different price tiers.

Table 4 SKU Inventory and Tier

Product Type	SKU	Unit Price	Orders	Tier Assigned
Smartphone	SKU1001	$20.75	1,972	Low
Smartphone	SKU1004	$766.80	1	Excluded (anomaly)
Smartphone	SKU1004	$791.19	2,017	Mid
Smartphone	SKU1005	$861.70	1	Excluded (anomaly)
Smartphone	SMP234	$1,139.68	1,987	High
Laptop	SKU1005	$463.96	2,011	Low
Laptop	LTP123	$674.32	1,960	High
Laptop	SKU1004	$932.03	1	Excluded (anomaly)
Smartwatch	SWT567	$459.50	1,980	Low
Smartwatch	SKU1003	$844.83	1,954	High
Tablet	SKU1002	$247.03	2,042	Low
Tablet	TBL345	$786.41	2,062	High
Headphones	HDP456	$361.18	2,010	Low (only tier)
Headphones	SKU1003	$826.04	1	Excluded (anomaly)

Table 5 Tier summary

Price Tier	High	Low	Mid
Headphones	0	2010	0
Laptop	1960	2011	0
Smartphone	1987	1972	2017
Smartwatch	1954	1980	0
Tablet	2062	2042	0

RQ2B Gender Analysis of Price Tier Preference

Chi-square test: Price Tier × Gender

For each product category with more than one tier, a chi-square test of independence was conducted to assess whether the distribution of price tier purchases differs between male and female customers. The null hypothesis is that the proportion of customers selecting each price tier is the same for both genders.

t-test: Mean Unit Price (Male vs Female)

An independent samples t-test was conducted for each product type to test whether the mean *Unit Price* paid differs between male and female customers. For *Headphones*, all customers paid the identical price of $361.18, so the t-test is not applicable.

Table 6 RQ2 Gender Statistical Summary

Product Type	Male Mean ($)	Female Mean ($)	Chi-sq	p (chi-sq)	t-stat	p (t-test)
Smartphone	$650.56	$655.11	1.243	0.537	-0.377	0.706
Laptop	$565.67	$569.94	1.557	0.212	-1.280	0.201
Smartwatch	$648.85	$653.08	0.431	0.512	-0.688	0.492
Tablet	$520.34	$515.65	0.276	0.599	0.556	0.578
Headphones	$361.18	$361.18	-	-	-	-

All chi-square tests produce p-values well above 0.05, and all t-tests are likewise non-significant. Cohen's w effect sizes are all below 0.02, which falls into the negligible category. This indicates not only that the results are statistically non-significant, but that the practical difference between groups is essentially zero.

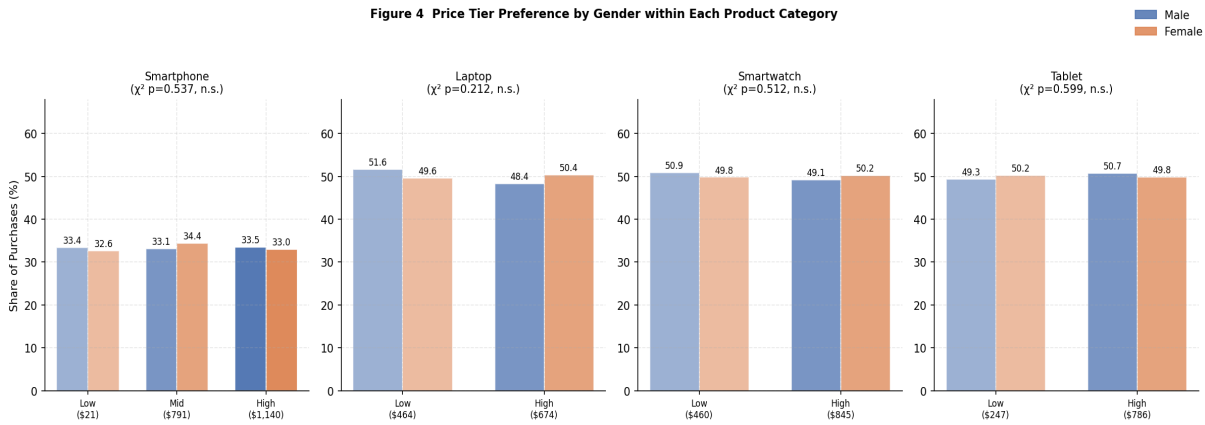

Figure 4 Price Tier Preference by Gender within Each Product Category

Figure 4 shows meaningful tier distributions. In every subplot, the male and female bars are nearly at the same height, visually confirming the chi-square results. The x-axis labels now show both the tier name and the actual *SKU* price, making the price structure transparent.

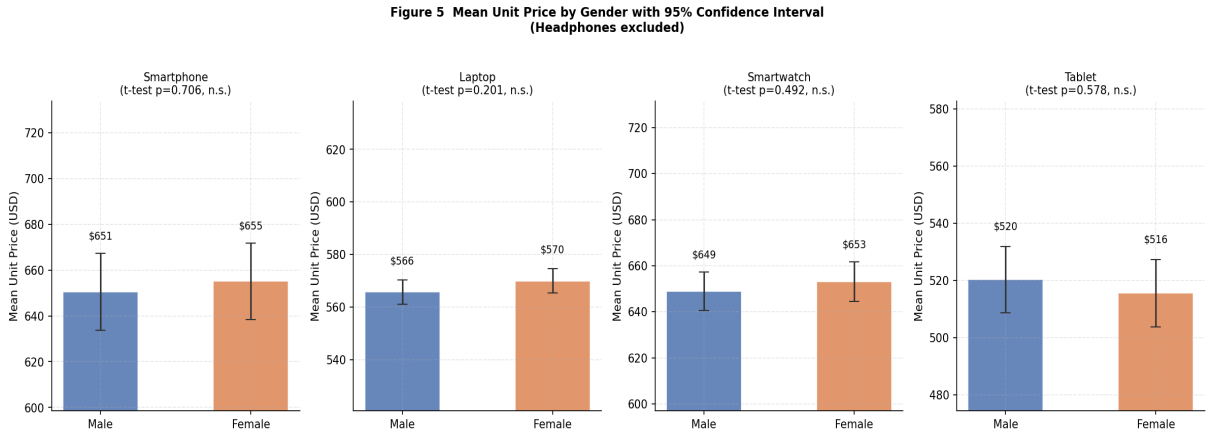

Figure 5 Mean Unit Price by Gender with 95% Confidence Interval

Figure 5 shows the mean *Unit Price* paid by male and female customers within each product category, with 95% confidence intervals. The error bars for male and female overlap substantially in every category, reinforcing the t-test findings. The dotted reference lines

indicate the actual *SKU* price boundaries within each category, illustrating that the average prices for both genders fall at almost identical positions within the price range.

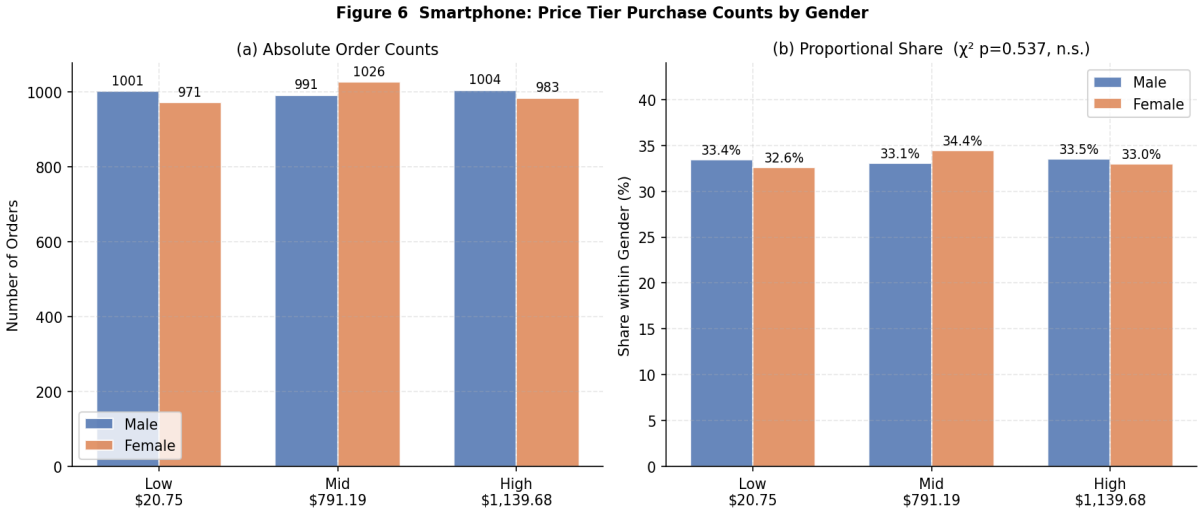

Figure 6 Smartphone: Price Tier Purchase Counts by Gender

Figure 6 provides a detailed view of the *Smartphone* category, the only product type with three distinct price tiers. *Figure 6(a)* shows the absolute order counts and confirms that order volumes are balanced across tiers for both genders. Each tier accounts for approximately 1,000 orders per gender. *Figure 6(b)* shows the proportional share within each gender, confirming that the distribution is essentially flat at approximately 33% per tier for both male and female customers (chi-square p = 0.537).

Across all four product categories where tier comparison is possible, the analysis consistently finds no statistically significant or practically meaningful gender difference in price tier preference. Male and female customers select entry-level, mid-range, and premium product models at virtually identical rates.

RQ2C Age Group Analysis of Price Tier Preference

The same analysis was extended to age group to test whether older or younger customers systematically prefer lower or higher priced product models. One-way ANOVA tests were used to assess mean unit price differences across age groups, and chi-square tests assessed tier distribution differences.

Table 7 Age Group Statistical Summary (One-way ANOVA)

Product Type	18-30 Mean ($)	31-45 Mean ($)	46-60 Mean ($)	61-80 Mean ($)	F-stat	p-value
Smartphone	$660.8	$658.8	$661.1	$637.2	1.057	0.366
Laptop	$566.5	$570.5	$564.3	$569.5	0.722	0.539
Smartwatch	$641.3	$653.8	$648.5	$656.4	1.122	0.339
Tablet	$526.6	$519.4	$523.0	$508.0	1.015	0.385
Headphones	$362.2	$361.2	$361.2	$361.2	1.190	0.312

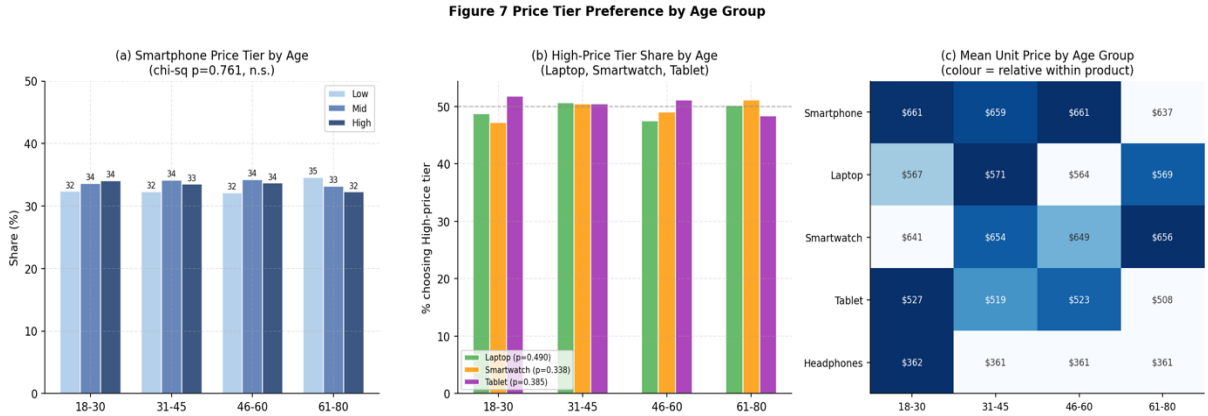

Figure 7 Price Tier Preference by Age Group

Table 7 and *Figure 7* show that age group is not a significant predictor of price tier preference in any product category. ANOVA F-statistics range from 0.72 to 1.19, with p-values between 0.312 and 0.539. For the *Smartphone* category, chi-square p = 0.761. The heatmap in *Figure 7(c)* illustrates that mean *unit price*s across age group are nearly uniform within each product category. Pearson correlations between *Age* (continuous) and *Unit Price* are all below |r| = 0.03 and are non-significant.

The convergent finding from RQ2B and RQ2C is that neither gender nor age group explains price tier selection.

RQ3 Revenue Drivers and Add-on Behaviour by Product Price Level
RQ3A Total Price Distribution and Product-Level Analysis

Table 8 Total Price Descriptive Stats by Product Type

Product Type	Mean	Std	Min	25%	50%	75%	Max
Headphones	$2,009.6	$1,044.2	$361.2	$1,083.5	$2,167.1	$2,889.4	$4,130.2
Laptop	$3,095.6	$1,779.4	$464.0	$1,391.9	$2,783.8	$4,175.6	$6,743.2
Smartphone	$3,599.3	$3,443.2	$20.8	$166.0	$3,164.8	$6,329.5	$11,396.8
Smartwatch	$3,567.9	$2,174.9	$459.5	$1,838.0	$3,216.5	$4,595.0	$8,448.3
Tablet	$2,853.8	$2,231.4	$247.0	$988.1	$2,223.3	$3,932.0	$7,864.1

As shown in *Table 8*, the mean *Total Price* is broadly similar across product types, ranging from approximately $2,009.6 for *Headphones* to $3,599.3 for *Smartphone*. However, *Smartphone* displays a notably high standard deviation of $3,443.2, meaning that individual transaction values are highly dispersed. This is explained by the fact that *Smartphone SKUs* span an exceptionally wide unit price range, from $20.75 to $1,139.68, combined with a purchase quantity range of 1 to 10 units, resulting in *Total Price* anywhere from $20.8 to $11,396.8.

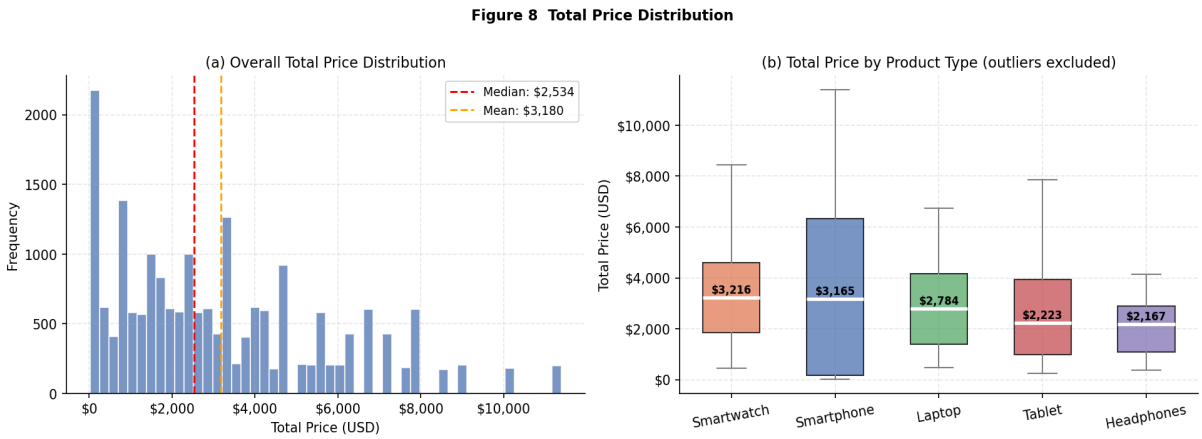

Figure 8 Total Price Distribution: Overall Histogram and Box Plot by Product Type

Figure 8(a) shows that the overall distribution of *Total Price* is right-skewed, with a long tail extending above $8,000. The median ($2,534) is substantially lower than the mean ($3,180), consistent with the presence of high-value outliers. *Figure 8(b)* compares distributions across product categories. *Headphones* show the most compact distribution, while *Smartphone* exhibits the greatest spread. *Smartwatch* and *Laptop* show moderate spreads with relatively symmetrical interquartile ranges.

RQ3B Quantity and Unit Price as Primary Revenue Drivers

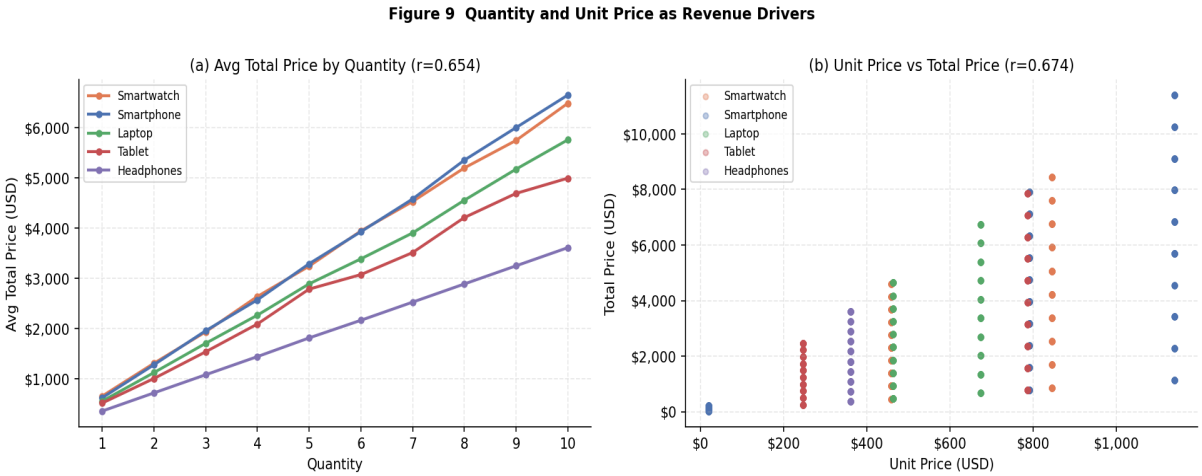

Figure 9 Quantity and Unit Price as Revenue Drivers

Figure 9 confirms that both *Quantity* and *Unit Price* are significant drivers of *Total Price*. The Pearson correlation coefficient between *Quantity* and *Total Price* is 0.654, while the correlation between *Unit Price* and *Total Price* is 0.674. As shown in *Figure 9(a)*, the relationship between *Quantity* and *Total Price* is approximately linear for all product types, with *Smartphone*, *Smartwatch*, and *Laptop* showing the steepest slopes, indicating that each additional unit purchased contributes $350-$570 to the total bill depending on product type. *Figure 9(b)* illustrates that higher *Unit Price* products are associated with higher *Total Price*, but the scatter is substantial, partly because *Total Price* is also driven by *Quantity*.

RQ3C Add-on Behaviour by Product Model Price Level

When I look at the figures of *Smartphone*, it hit me with a more specific question:

Do customers who purchase lower-priced product models differ in their add-on purchasing behaviour compared to customers who purchase higher-priced models of the same product type?

To investigate this, the dataset was analysed at the *SKU* level, since each product type contains two to four distinct *SKUs* with fixed *Unit Prices*.

Table 9 SKU-Level Add-on Analysis

Product Type	SKU	Unit Price	Orders	Attach Rate (%)	Avg Add-on ($)
Smartphone	SKU1001	$20.75	1,972	74.7%	$41.16
Smartphone	SKU1004/791	$791.19	2,017	74.8%	$41.16
Smartphone	SMP234	$1,139.68	1,987	75.8%	$83.89
Laptop	SKU1005	$463.96	2,011	75.2%	$41.79
Laptop	LTP123	$674.32	1,960	76.0%	$84.66
Smartwatch	SWT567	$459.50	1,980	74.9%	$80.83
Smartwatch	SKU1003	$844.83	1,954	77.2%	$41.64
Tablet	SKU1002	$247.03	2,042	75.2%	$42.24
Tablet	TBL345	$786.41	2,062	76.7%	$82.44
Headphones	HDP456	$361.18	2,010	76.0%	$82.64

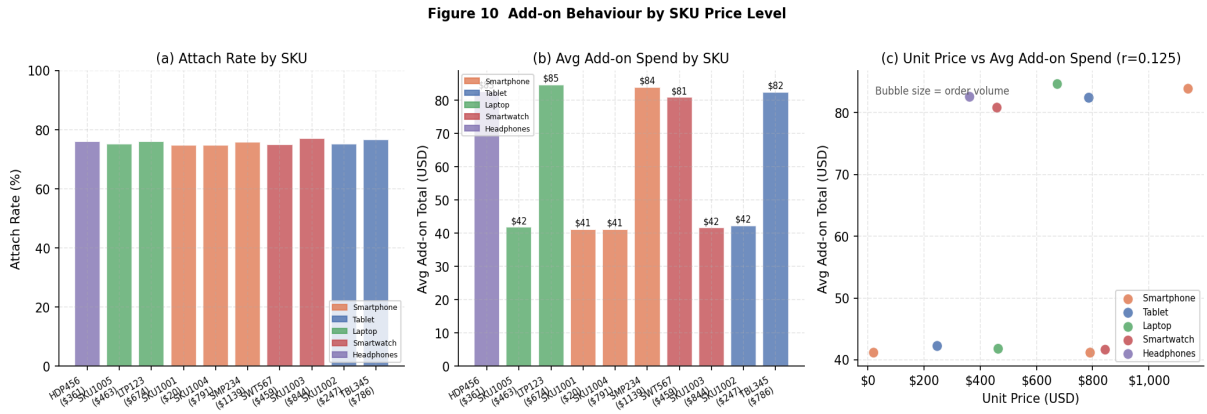

Figure 10 Add-on Behaviour by SKU: Attach Rate, Average Add-on Spend, and Unit Price vs Add-on Spend

The *SKU* level analysis reveals an important distinction between two dimensions of add-on behaviour. As shown in *Figure 10(a)* and *Table 9*, the add-on attach rate (the proportion of orders that include at least one add-on) is remarkably consistent across *SKUs*, ranging between 74.7% and 77.2% regardless of *Unit Price*. The chi-square test for *Smartphone* confirms that the difference in attach rate between low-price and high-price models is not statistically significant ($p = 0.385$).

However, the average add-on spend per order tells a different story. As shown in *Figure 10(b)*, *SKUs* with lower unit prices (e.g., *Smartphone SKU1001* at $20.75, *Laptop SKU1005* at $463.96) are associated with average add-on totals of approximately $41-$42, while *SKUs* with higher *Unit Price* (e.g., *Smartphone SMP234* at $1,139.68, *Laptop LTP123* at $674.32) are associated with average add-on totals of approximately $81-$85. Independent samples t-tests confirm that these differences are statistically significant for both *Smartphone* ($p < 0.001$) and *Laptop* ($p < 0.001$). This pattern aligns with the broader literature on willingness-to-pay, which suggests that consumers who invest in premium products tend to exhibit higher overall expenditure on associated accessories and enhancements (Ashraf et al., 2017).

Figure 10(c) provides a visual summary of this pattern. The scatter plot of *Unit Price* against average *Add-on Total* shows that the relationship is not monotonically increasing, but rather that add-on pricing appears to cluster around two distinct levels: approximately $41 and

approximately $83. This suggests that add-on products are priced in tiers that correspond to the main product price level, rather than being offered at uniform prices across all models.

RQ3D Add-on Type Composition and Overall Revenue Contribution

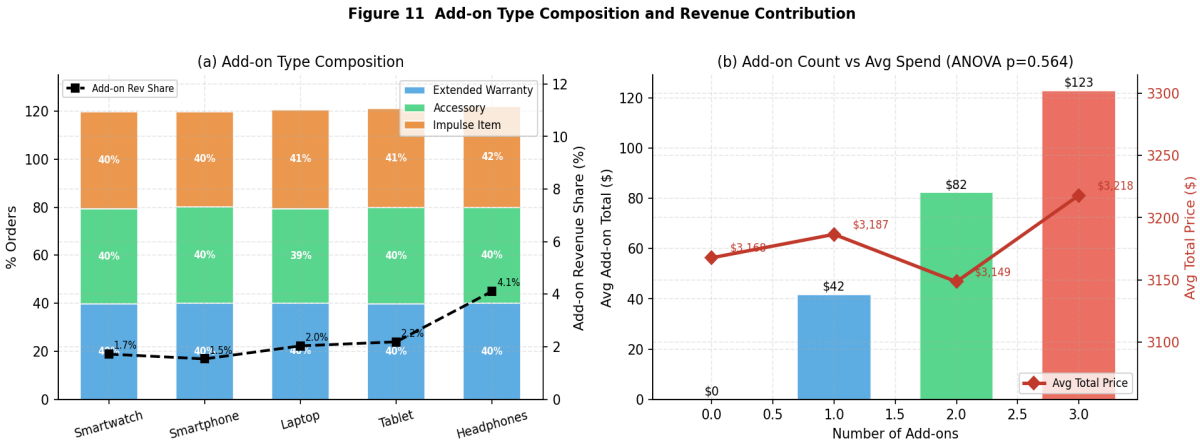

Figure 11 Add-on Type Composition and Revenue Contribution by Product Category

Figure 11(a) shows that the three add-on types (Extended Warranty, Accessory, and Impulse Item) are distributed almost equally across all product categories, each accounting for approximately 40% of orders within each product type. This uniformity suggests that the availability and pricing of add-on types are standardised across the product catalogue.

The line in *Figure 11(a)* shows the add-on revenue share as a percentage of total revenue per product type. *Headphones* have the highest add-on revenue share at 4.1%, followed by *Tablet* at 2.2% and *Laptop* at 2.0%. *Smartphone* have the lowest share at 1.5%. This pattern is largely explained by the inverse relationship between base product price and relative add-on contribution. For *Headphones*, whose base *Unit Price* is the lowest among all categories ($361.18), add-on purchases represent a proportionally larger fraction of total revenue.

Figure 11(b) confirms that each additional add-on item contributes approximately $42 in average add-on total. Despite this consistent unit contribution, the one-way ANOVA test for the effect of add-on count on *Total Price* is not statistically significant ($F = 0.68$, $p = 0.564$). This is expected given that add-ons account for only 2.0% of total revenue ($1,244,897 out of $63,601,994), and the dominant variance in *Total Price* is driven by *Unit Price* and *Quantity* differences.

Table 10 Revenue Summary by Product Type

Product Type	Orders	Avg Total ($)	Median Total ($)	Avg Unit ($)	Avg Qty	Add-on Rev (%)
Smartphone	5,978	$3,599	$3,165	$653	5.46	1.54%
Smartwatch	3,934	$3,568	$3,217	$651	5.50	1.72%
Laptop	3,972	$3,096	$2,784	$568	5.43	2.03%
Tablet	4,104	$2,854	$2,223	$518	5.52	2.19%
Headphones	2,011	$2,010	$2,167	$361	5.56	4.11%

Conclusions

Three consumer behavior-related questions were investigated in this article using a dataset of 19,999 electronic product sales transactions from September 2023 to September 2024.

For RQ1, no statistically significant differences were found across any of the four behavioural indicators: loyalty conversion, order cancellation, credit card usage, and shipping type, when analysed by either *Gender* or *Age* group. The range of *Gender* p-values is 0.171 to 0.654. P-values for *Age Groups* vary from 0.219 to 0.847. A more urgent operational concern than any differential based on demographics is the high total cancellation rate of about 33%. Rather than an underlying behavioural difference, the peak conversion time difference between male (August 2024) and female (June 2024) clients might indicate a delayed reaction to the same marketing stimuli.

Price tier preference for RQ2 is not substantially correlated with age or gender in any product category. Using SKU-rank-based tiers, corrected chi-square tests produce p-values of 0.212-0.599 for *Gender* and 0.339-0.761 for *Age Groups*. F-statistics of 0.72-1.19 (all non-significant) are obtained from a one-way ANOVA for mean unit price by age group. Throughout, Cohen's w and effect size measures are negligible.

Therefore, RQ1 and RQ2 consistently show that age and gender are not useful segmentation factors for this clientele. This may be due to the homogenising effect of personalised recommendation systems (Hallikainen et al., 2022) and is generally consistent with a systematic review revealing that demographic disparities in e-commerce behaviour are context-dependent and frequently lower than predicted (Kanwal et al., 2022).

For RQ3, the analysis identified *Quantity* and *Unit Price* as the two primary drivers of *Total Price*, with Pearson correlations of 0.654 and 0.674 respectively. Add-on purchases, while present in approximately 75-76% of all orders, contribute only 2.0% to *Total Revenue*. The *SKU* level analysis revealed a particularly notable finding: the add-on attach rate is consistent across model price levels within the same product category (e.g., 74.7% for the $20.75 *Smartphone* model and 75.8% for the $1,139.68 *Smartphone* model), but the average add-on spend is significantly higher for customers who purchase more expensive models. For *Smartphone*, customers purchasing the premium model spent an average of $83.89 on add-ons, compared with $41.16 for customers purchasing the entry-level model ($p < 0.001$). This pattern aligns with evidence that consumers who commit to premium products tend to exhibit a higher willingness to pay for complementary items (Ashraf et al., 2017) and that tiered product structures can effectively capture heterogeneous consumer valuations across different spending segments (Jedidi et al., 2003).

From a business perspective, the absence of gender-based and age-based price tier preferences suggests that tier-differentiated marketing campaigns targeted at one gender or a certain age group are unlikely to yield incremental returns in this market. Pricing and product assortment decisions should be driven by factors other than customer gender or age. Based on the RQ3 findings, the company can adopt a tiered pricing strategy: entry-level products can be priced relatively low to attract a broad customer base, while premium products can be priced significantly higher. Offering a variety of add-ons provides an additional revenue lever.

Real-world examples include Apple, which sells entry-level and premium devices with extensive accessory ecosystems, and Kindle e-readers, which offer different models with optional add-ons such as covers and storage expansions.

Furthermore, the high cancellation rate of approximately 33% warrants operational attention. Improving shipping speed and flexibility, and investing in loyalty retention programmes, could help convert the large *Never Member* group and reduce churn among existing members.

Limitations

Several limitations should be considered when interpreting the findings of this report.

First, the dataset records *Total Price* as *Unit Price* multiplied by *Quantity*, with *Add-on Total* billed separately and not included in *Total Price*. While this is confirmed by the data structure, the commercial relationship between main product pricing and add-on pricing cannot be fully established from the dataset alone. It is not clear whether higher add-on spending by premium model purchasers reflects deliberate tiered pricing of add-on products by the retailer, or whether it reflects greater willingness-to-pay among customers who self-select into purchasing higher-priced models (Ashraf et al., 2017).

Second, the demographic analyses rely on two variables: gender and age, which may carry measurement limitations. Gender is recorded as a binary variable, which may not capture the full spectrum of customer identities. Age is recorded in whole years, and the four-bracket grouping, while consistent with the data's quartile structure, is necessarily arbitrary. Furthermore, if gender or age data is inferred rather than self-reported, measurement error may attenuate any true demographic effects (Kanwal et al., 2022). It should also be noted that the original quantile-based binning approach for RQ2 produced a degenerate result. The corrected SKU-rank method yields substantively valid results, though the limited number of distinct SKUs per category (two to four) constrains the granularity of the tier analysis.

Third, the dataset contains only 10 distinct *SKUs* across five product categories, with two to four models per category. The price variation within categories is limited to a small number of discrete price points. The observed clustering of add-on spend around two levels (approximately $41 and $83) may be a product of this coarse *SKU* structure rather than a reflection of broader pricing dynamics. A richer dataset with more *SKU* variation would allow a more nuanced examination of the relationship between product price and add-on spending.

Finally, the analysis is cross-sectional and covers a single year. It is not possible to determine whether the observed patterns are stable over time or whether they reflect seasonal or promotional effects. The monthly conversion trend analysis in RQ1 suggests that marketing campaigns may drive short-term behavioural changes, but the dataset does not include information on promotions, discounts, or campaign activities that would be necessary to test this interpretation directly.

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